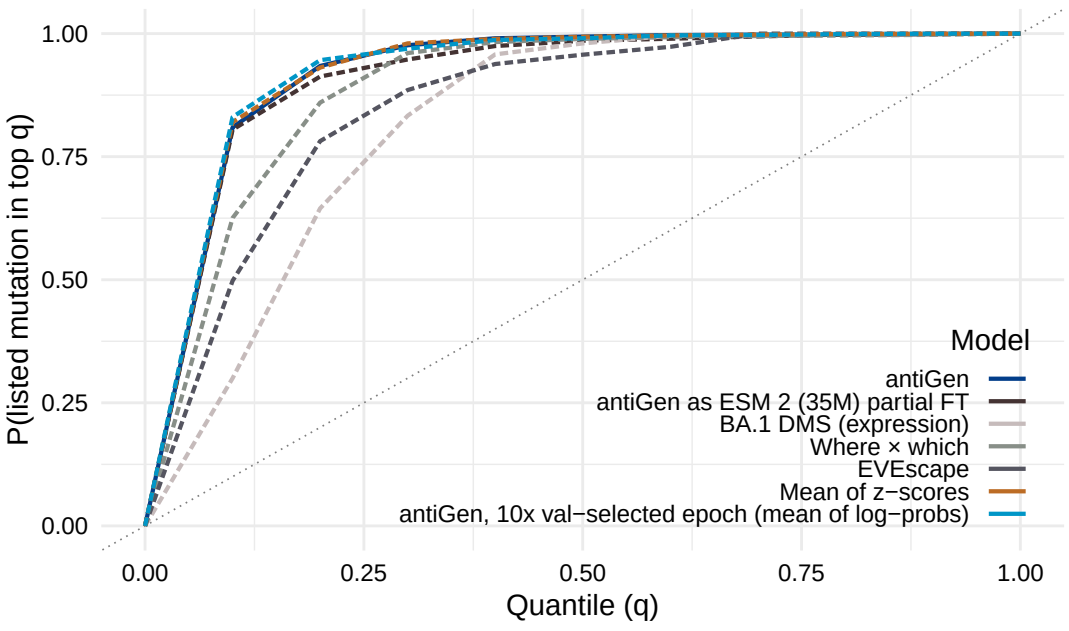
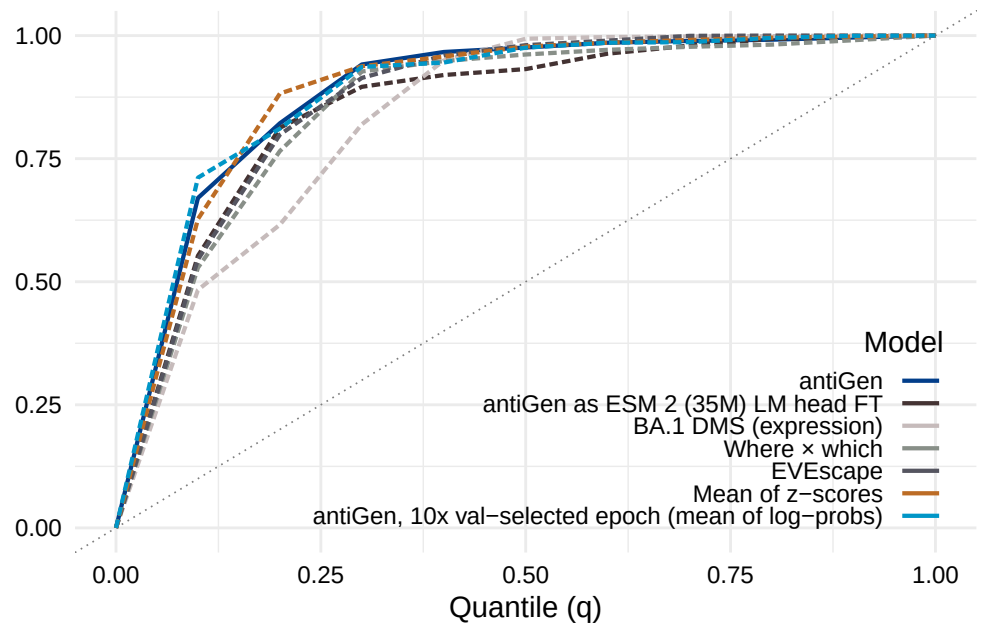
Model type: antiGen; antiGen Finetunes; Deep Mutational Scans; Pandemic Data in Training Window; Pre−Pandemic Data Only; Ensembles; Val−selected, averaged over 10 holdouts

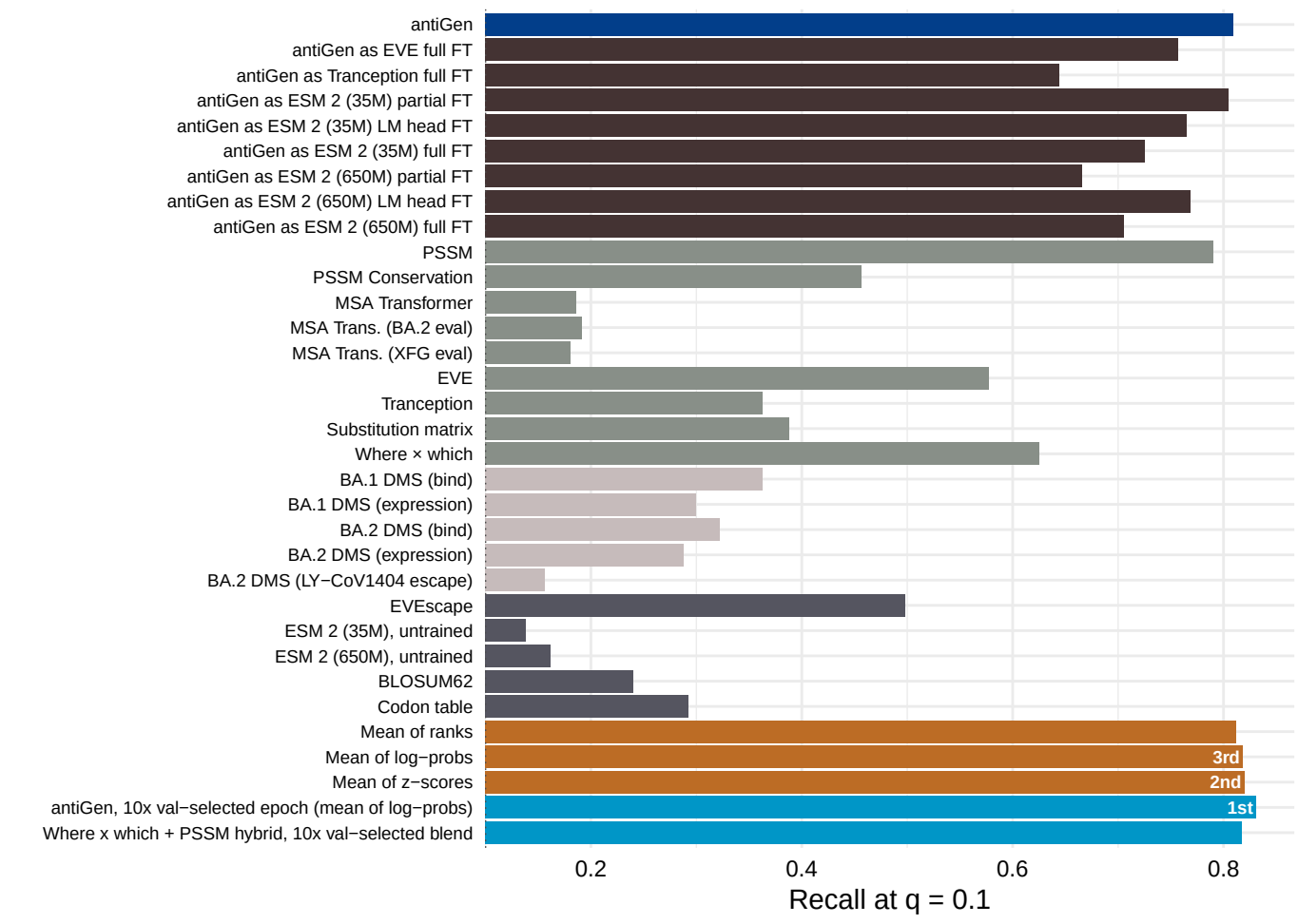
**A** Recall, all list cells, event−weighted [20 AA, revised corpus]



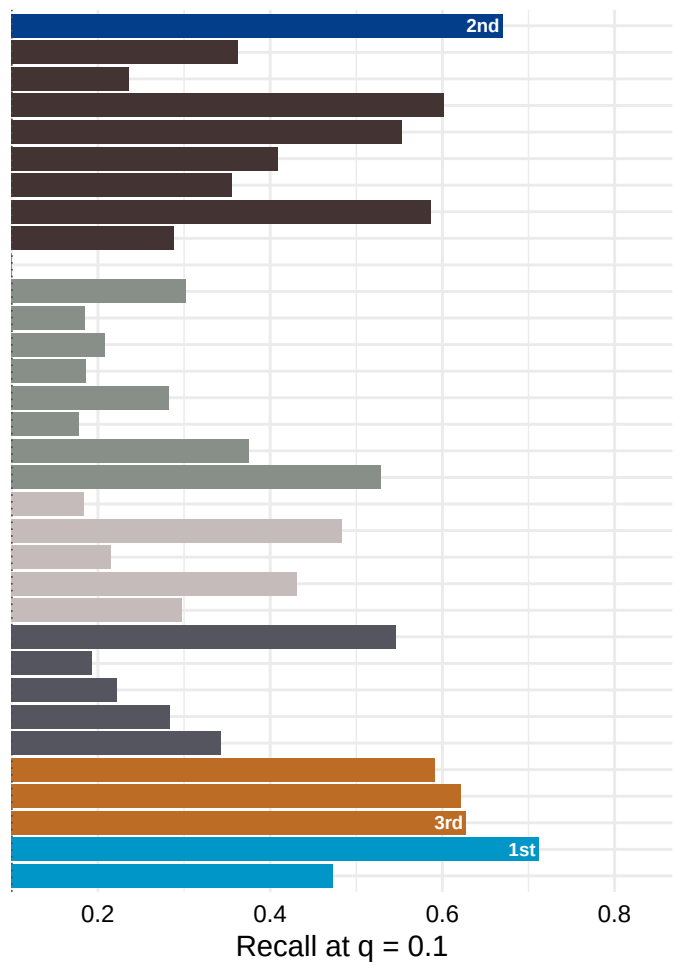

**B** Recall, de novo, event−weighted [20 AA, revised corpus]



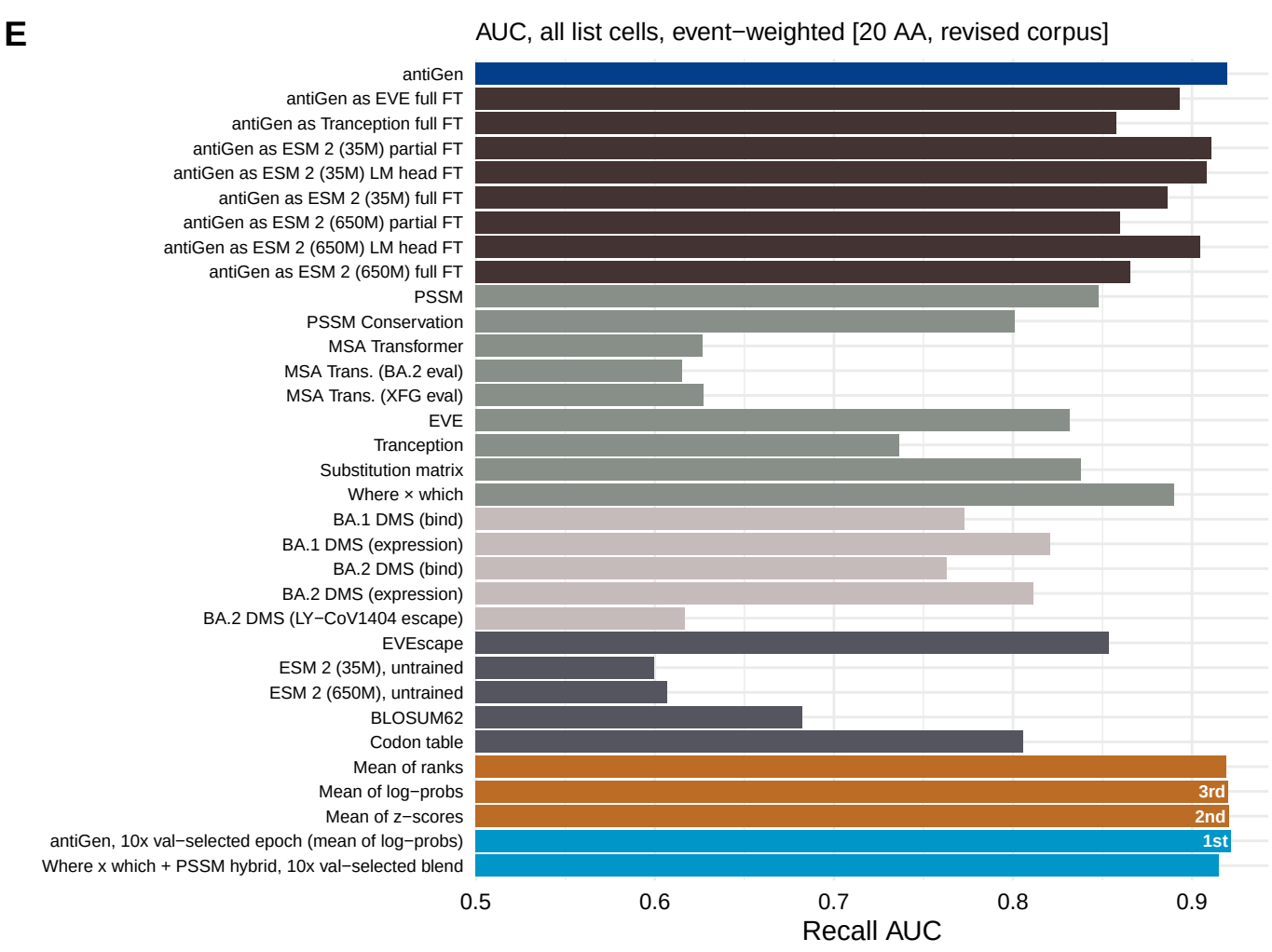

**C** Top decile, all list cells, event−weighted [20 AA, revised corpus]



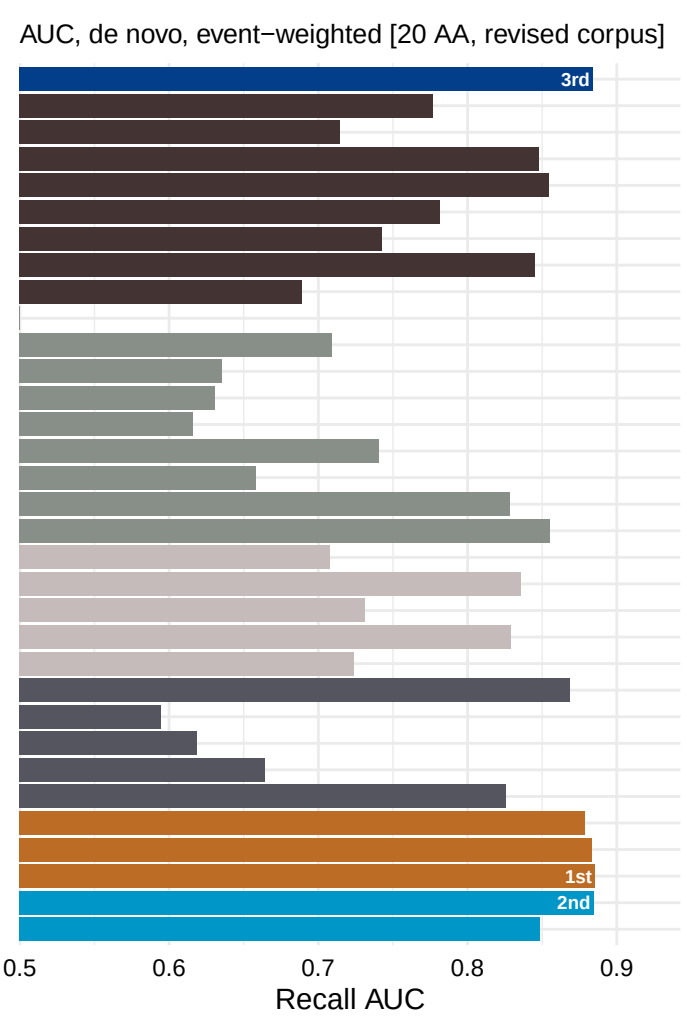

**D** Top decile, de novo, event−weighted [20 AA, revised corp



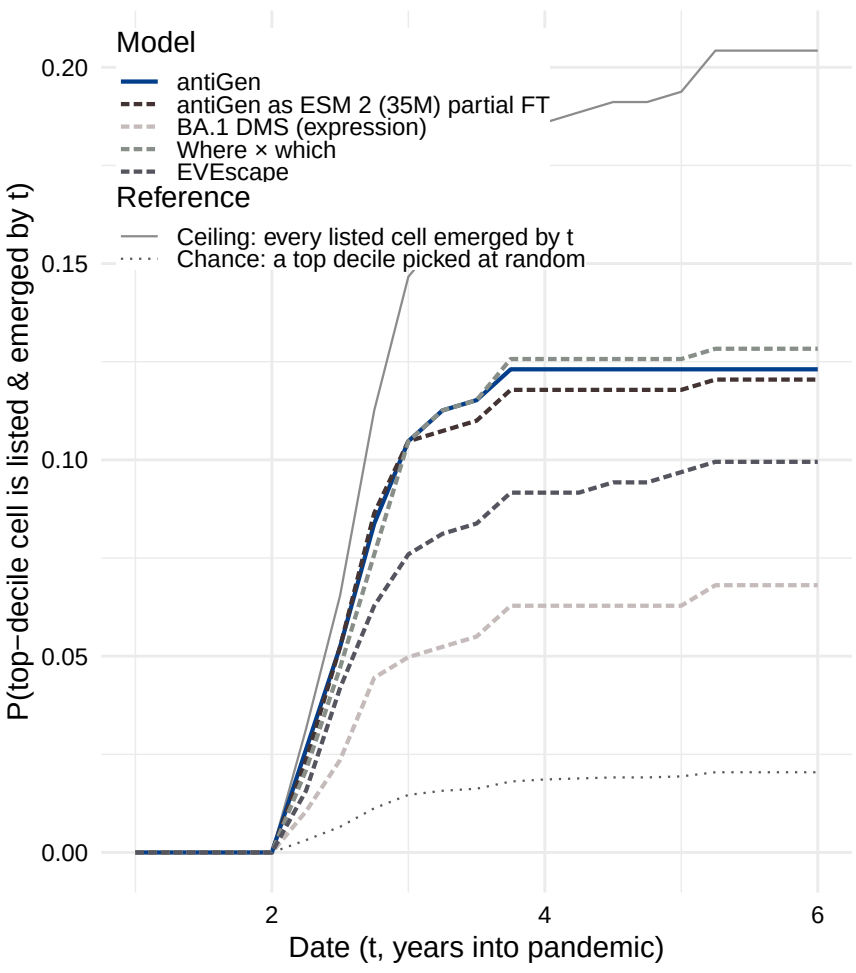

**E** AUC, all list cells, event−weighted [20 AA, revised corpus]



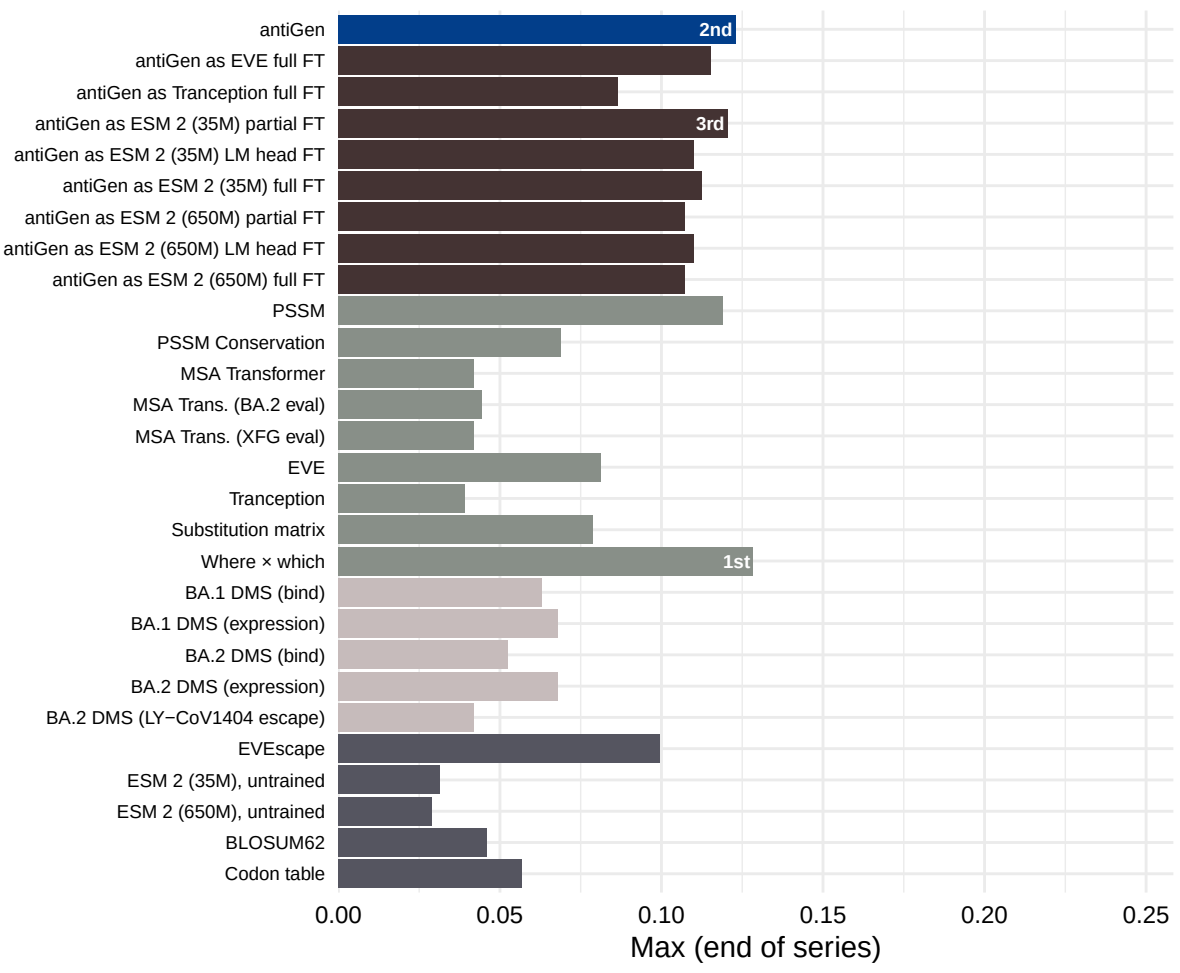

**F** AUC, de novo, event−weighted [20 AA, revised corpus]



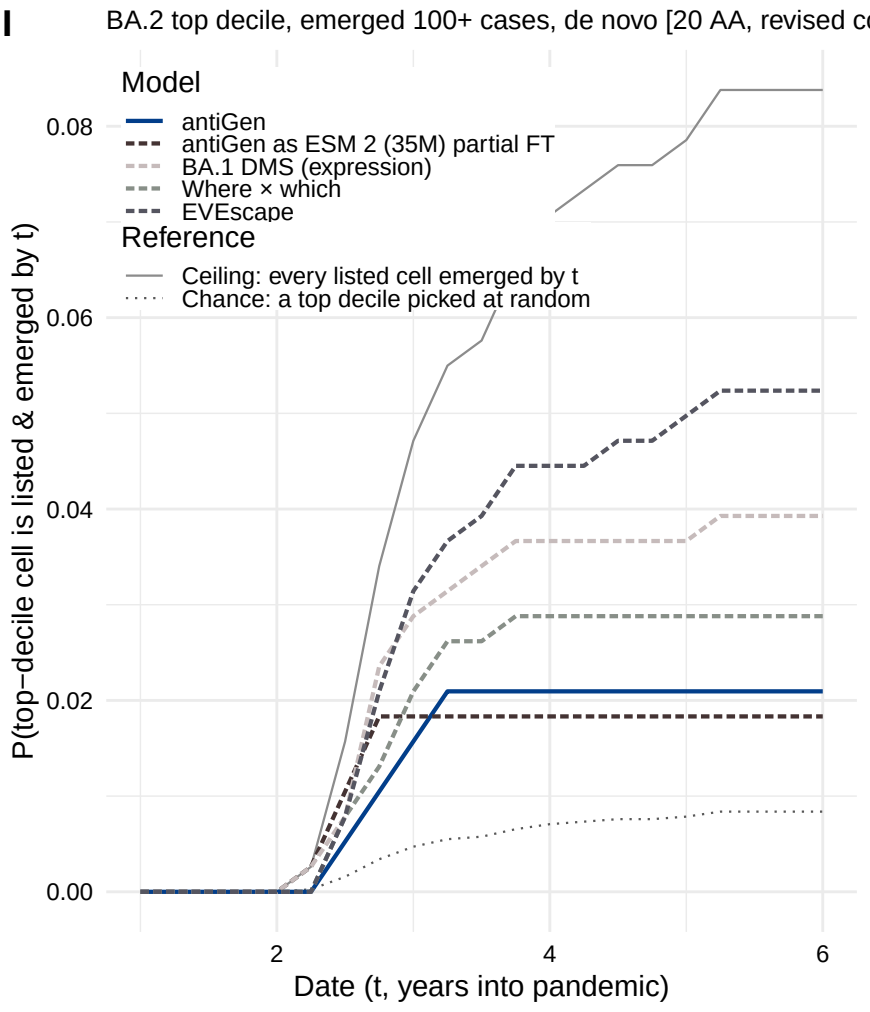

**G** BA.2 top decile, emerged 100+ cases, all list cells [20 AA, revised

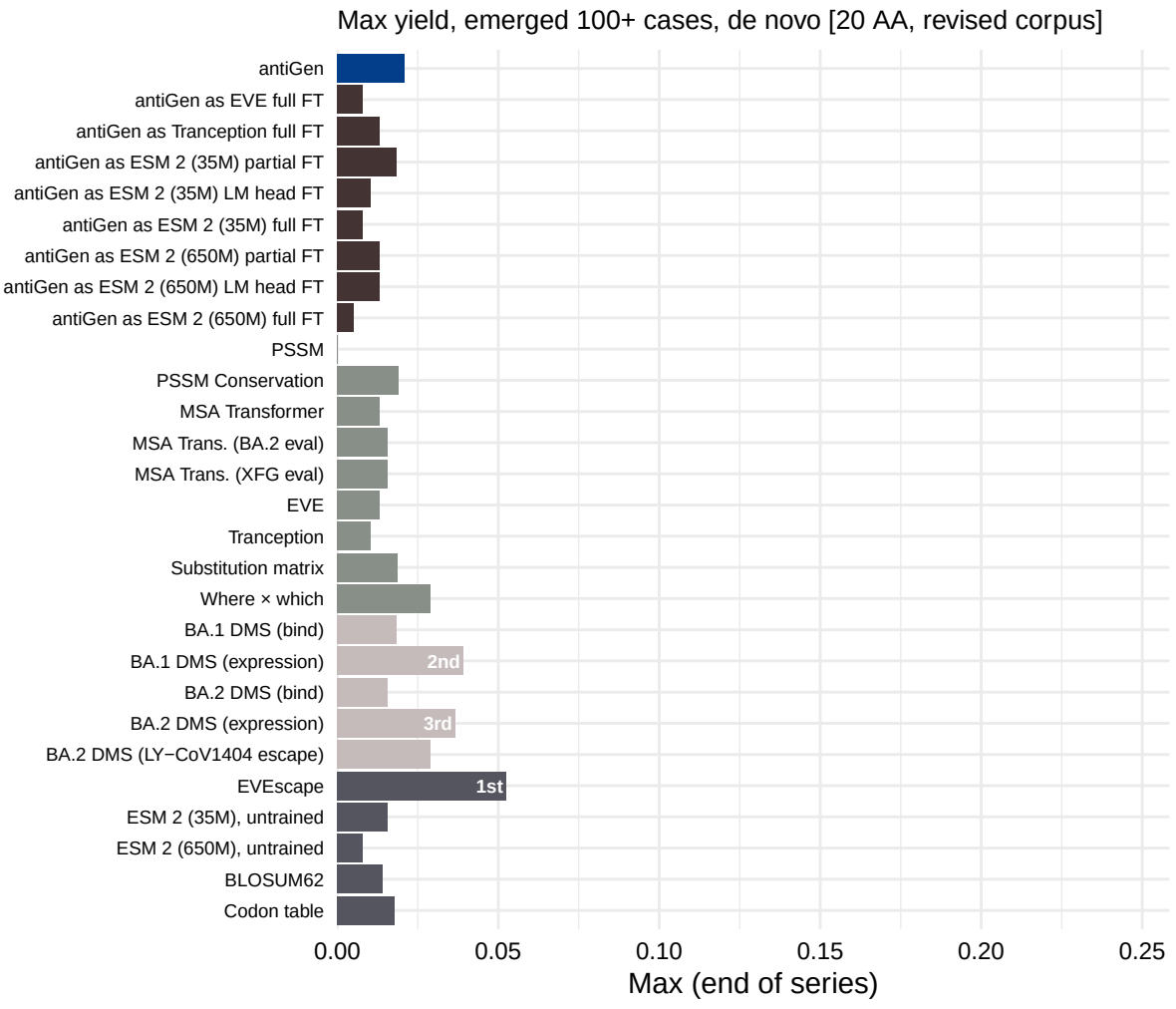



**H** Max yield, emerged 100+ cases, all list cells [20 AA, revised corpus]




**I** BA.2 top decile, emerged 100+ cases, de novo [20 AA, revised c




**J** Max yield, emerged 100+ cases, de novo [20 AA, revised corpus]